

Función diferenciable

Definición 1 (Diferenciabilidad en un punto). Sea $U \subset \mathbb{R}^n$ abierto, $f: U \rightarrow \mathbb{R}^m$ es diferenciable en $x_0 \in U$

$$\iff \exists T: \mathbb{R}^n \rightarrow \mathbb{R}^m \text{ lineal} : \exists \lim_{h \rightarrow 0} \frac{\|f(x_0 + h) - f(x_0) - T(h)\|}{\|h\|} = 0.$$

Definición 2 (Diferenciabilidad). Sea $U \subset \mathbb{R}^n$ abierto, $f: U \rightarrow \mathbb{R}^m$ es diferenciable

$$\iff \forall x_0 \in U : f \text{ es diferenciable en } x_0.$$

Referenciado en

- teo-densidad-induce-metrica
- gradiente
- lem-directamente-conforme-imp-holomorfa
- teo-fn-inversa-holomorfes
- teo-derivacion-bajo-el-signo-integral
- teo-cauchy-riemann
- apl-conforme
- cor-convolucion-regularidad

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